

Dictionary data type:

Dictionary is the collection of the key value pairs enclosed between the pair of curly brackets.

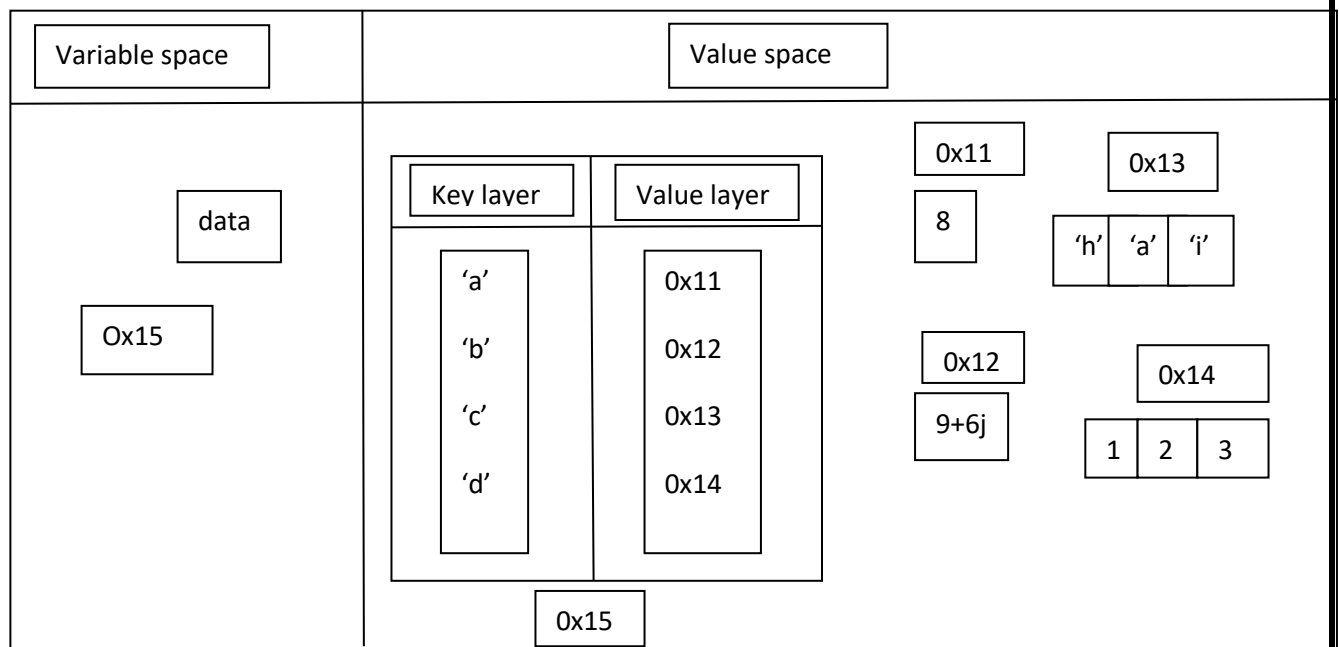
Syntax:

Var = {K1:V1, K2:V2,..... Kn:Vn}

- Where the key value pair is separated by the comma (',') operator.
- The key and value are separated by colon (':').
- In dictionary we cannot store duplicate keys, if we store then the previous value will get over ridden by the next value.
- In dictionary key should be 'immutable'.
- The default value of the dictionary data type is { }, which is internally equal to Boolean False.

Memory allocation of dictionary data type:

Data = {'a':8, 'b':9+6j, 'c':'hai':'hello', 'd':[1,2,3]}



- Indexing is not possible in dictionary but keys will act as indexing internally.
- The syntax used to get the value based on keys is

`Variable[key]`

- The syntax to create new key value pair is

`Variable[key] = new_value`

Attributes of dictionary data type:

1. **Keys ()**: it is function, which is used to keys all the keys present in the dictionary. It returns the output in the form of list.
`Variable.keys()`
2. **Values ()**: it is function, which is used to values all the values present in the dictionary. It returns the output in the form of list.
`Variable.values()`
3. **Items ()**: it is a function which is used to get all the key value pairs. It returns the output in the form of list of tuples.
`Variable.items()`
4. **Pop ()**: it is a function which is used to delete the particular entry from the collection.
`Variable.pop(value)`
5. **Popitem()**: it is a function which is used to delete the last entry of the collection.
`Variable.popitem()`
6. **Copy ()**: it is a function which is used to copy the entire collection from one variable to another variable.
`Variable.copy()`
7. **Clear ()**: it is a function which is used to clear the entire dictionary.
`Variable.clear()`
8. **Get ()**: it is a function which is used get the value of the specified key.
`Variable.get(key)`

9. **Update ()**: it is a function which is used to update the existing dictionary with the specific entry.

Variable.update({key:value})

Since we can modify the value original collection, dictionary is considered as a “mutable data type”.

Slicing:

Slicing is a phenomenon of extracting a group of data item from the original collection.

Syntax:

Variable[start_index : end_index +1 : updation] → left to right

Variable[start_index : end_index -1 : updation] → right to left